·综 述·

基于大脑皮层不同刺激靶点探讨*Theta*爆发式经颅磁刺激对卒中后失语症的影响

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摘要 卒中后失语症(PSA)的患病人群日益递增,此类患者临床表现为不同程度言语功能障碍,言语能力作为人类特有的核心认知与社交功能,其重要性贯穿于身心健康、文化延续等多个维度,但该病发生与恢复的脑机制不明,传统治疗效果欠佳,促使研究人员不断探索新的治疗方法。经颅磁刺激(TMS)是一种无创、安全的非侵入性大脑刺激技术,*Theta*爆发式经颅磁刺激(TBS)作为特殊TMS模式,具有耗时短、强度低、效应强、毒副作用少等优势,近年来临床应用渐多。本文介绍了TBS的特点、常用参数及安全性,值得注意的是不同参数治疗PSA患者存在效果差异,需要更多的研究来明确最佳参数。现有的研究提示其治疗PSA的机制包括调节大脑代谢、改善功能网络连接、促进半球间兴奋性平衡等,目前国内外研究结果表明TBS作用于大脑皮层,如额下回、初级运动皮层、Wernicke区和额叶背外侧等脑区,分别对自发言语、听理解、复述、命名等言语功能亚项的一项或几项功能有改善作用,表明TBS对PSA患者的言语功能恢复具有潜在的临床意义。此外,双侧或单侧多靶点TBS刺激结合言语训练的治疗方案,可实现功能互补,且TBS耗时短,多靶点刺激可行性较高。目前研究存在以短期效果为主、长期数据缺乏的局限,未来需探索小脑等更多潜在靶点,结合神经心理学量表与影像学技术,评估长期疗效并深入探讨脑机制,为PSA精准治疗提供依据。

关键词 Theta爆发式经颅磁刺激;卒中后失语症;刺激靶点;疗效;言语功能

脑卒中是一种高发病率、高致残率的疾病[1-2],其中约1/3脑卒中患者会出现卒中后失语症(post-stroke aphasia,PSA),表现为听理解、自发表达、复述、命名、阅读、书写等全部或部分功能不同程度障碍[3]。约50% PSA患者在脑卒中1年后仍有持续的交流障碍,严重影响其日常生活质量[4-5]。随着我国人口老龄化的趋势不断加剧[6],PSA罹患人群日益递增,然而PSA发生与恢复的脑机制尚未明确,言语训练、药物治疗等传统方法对PSA的治疗效果不佳。经颅磁刺激(transcranial magnetic stimulation,TMS)可透过颅骨直接刺激大脑皮层改变神经组织的兴奋性,是一种无创、无痛、易操作、安全性较高的非侵入性大脑刺激(non-invasive brain stimulation,NIBS),近年来成为康复科、精神心理科等学科的研究热点[7-8]。已有的研究证实虽然TMS可以改善PSA患者的言语功能,但不同TMS刺激模式及刺激靶点等参数对PSA患者言语功能的改善效果不尽相同[9-11]。*Theta*爆发式经颅磁刺激(*Theta* burst tran-

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scranial magnetic stimulation，TBS）与其他模式TMS相比，具有耗时短、强度低、效应强、性价比高等特点，通过短时间内高频刺激语言相关皮层改变神经元的活动模式，毒副作用的发生率较其他模式TMS低，临床的应用逐渐增多[12-14]。目前国内外TBS治疗PSA的研究采用了不同靶点刺激方案，对言语功能的改善存在差异。本文就TBS治疗PSA刺激靶点、疗效方面的研究综述如下，为今后临床研究提供一定参考。

1　TBS治疗PSA概述

1.1　TBS及其特点

1985年TMS开始在研究中使用，它是一种神经调控技术，通过在线圈中变化电流产生磁场，穿透颅骨直接作用于中枢神经系统，改变大脑皮层神经细胞的膜电位，诱导产生感应电流，激活神经通路，影响脑代谢和神经电活动，根据TMS刺激脉冲个数及规律不同，通常分为以下几种刺激模式：单脉冲TMS（single-pulse，sTMS）、双脉冲TMS（paired-pulse TMS，ppTMS）、重复经颅磁刺激（repetitive TMS，rTMS）以及TBS[15]。

TBS是一种特殊的TMS形式，它模拟大脑海马体和皮质回路活动中的内源性*Theta*波（频率为5 Hz左右），通过在短时间内提供丛状节律式的脉冲刺激，从而引起大脑皮质较长时间的兴奋性改变[16]。根据刺激与间歇时间的不同，以及诱导皮层兴奋性的不同，TBS主要分以下2种类型：① 诱发皮质长时程增强效应产生兴奋效应的间歇性TBS（intermittent TBS，iTBS）；② 诱发皮质长时程抑制效应产生抑制作用的连续性TBS（continuous TBS，cTBS）[15,17]。

与传统的rTMS相比，TBS具有刺激时间短、生理效应长的优势，传统rTMS单次治疗时长为20 min左右，而TBS仅需40～190 s，患者在TBS刺激过程中需要保持静止的时间更短，这种时间上的缩短不仅提高了治疗的效率，还增加了患者的舒适度和依从性[18]。同时TBS还具有刺激强度低的优势[19]，并能在一定程度上减少毒副作用的发生[14]。

在严格遵循TBS适应证及禁忌证下的TBS治疗是比较安全的[20]，较少发生癫痫发作等严重不良反应，但可能有轻微毒副作用，如头痛、头皮不适、面部肌肉抽搐、头晕、恶心、轻度疲劳。这些通常都是暂时的，随着治疗进行，患者会逐渐适应，毒副作用也会减轻[21]。一旦出现不良反应，应及时对症处理并严格观察。故TBS首次治疗前应进行脑电图检查，识别潜在的癫痫放电或其他异常脑电活动，尽量避免在TBS治疗过程中诱发癫痫发作或其他不良事件。

1.2　TBS治疗PSA的机制

目前大多数的研究认为PSA患者言语功能的恢复可能涉及以下几种机制：① 半球内代偿：左侧半球病变区及病变周围区的语言相关功能区重建；② 半球间代偿：在左侧半球语言功能区域受损后，右侧半球镜像区进行激活、重组；③ 半球间失代偿：右侧半球镜像区的不良激活干扰言语功能恢复[22-23]。TBS治疗PSA的机制与TMS类似，包括以下几个方面：① 在促进大脑代谢方面，TBS可能通过调节大脑的代谢活动来促进神经修复和言语功能恢复。通过刺激语言处理相关的区域，调节代谢产物（如N-乙型半胱氨酸、胆碱、肌醇）的释放、调节皮层兴奋性及促进血清脑源性神经营养因子（brain-derived neurotrophin factor，BDNF）相关基因的表达等，促进神经可塑性，从而有助于恢复语言功能[24-26]。② 在功能网络连接方面，PSA通常与大脑语言网络中的功能连接中断有关，TBS通过增强或重新配置受损区域与未受损区域之间的连接，提高PSA患者的言语功能[27]。③ 半球间抑制学说构成了大多数使用TBS作为治疗PSA患者研究的基础[28]。在正常生理情况下，人体左右大脑半球经胼胝体相互抑制，处于动态平衡状态[29]，卒中破坏了左右半球之间的抑制平衡，左半球对右半球的抑制减弱，右半球镜像区不良激活，其兴奋性增加，进而增加对左半球的抑制，导致左半球病变区兴奋性更加降低[30]。TBS治疗促进左右半球兴奋性不平衡状态正常化[31-32]。

1.3　TBS治疗PSA的刺激参数

根据胼胝体相互抑制原理，PSA左侧大脑半球的TBS刺激常选用iTBS，刺激强度为80% 运动阈值（active motor threshod，AMT），丛内频率50 Hz，丛内计数3，丛内刺激时间0.06 s，丛间频率5 Hz，丛间计数10，刺激时间2 s，间歇时间8 s，重复次数20，共计600个脉冲；PSA右侧大脑半球的TBS刺激常选用抑制性的cTBS，刺激强度为80% AMT，丛内频率50 Hz，丛内计数3，丛内刺激时间0.06 s，丛间频率5 Hz，丛间计数200，刺激时间40 s，间歇时间8 s，共计600个脉冲[17]。

TBS不同刺激参数与上述刺激参数的效果可

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能略有差异。GAMBOA等[33]研究发现虽然标准的600个脉冲iTBS对运动皮层兴奋性有促进作用,但1 200个脉冲的延长iTBS方案会产生相反的结果——转化为具有抑制性,而通常抑制性的cTBS在刺激持续时间增加一倍时转化为具有促进性,皮质兴奋性将反向抑制。这提示临床上制定治疗方案时不能简单地通过延长TBS来故意增强TBS诱导的可塑性。因此,还需要更多的临床研究以获得确切统一的TBS治疗PSA的刺激参数。

2　基于大脑皮层不同刺激靶点的TBS应用于PSA的国内外研究现状

2.1　TBS作用于PSA患者左侧大脑半球的研究

1项荟萃分析表明右侧大脑半球低频rTMS(通常指刺激频率≤1 Hz)方案被公认在治疗PSA中显示出较为确定的疗效[34]。高频rTMS(通常指刺激频率>1 Hz)方案应用于左侧大脑半球还是右侧大脑半球仍有争议[35-36]。在TBS治疗PSA的研究中较多采用iTBS方案刺激左侧大脑半球,并且结果显示左侧大脑半球iTBS对PSA患者言语功能的恢复有利,但由于iTBS刺激靶点不同,对言语功能的改善也存在差异[16]。

2.1.1　左额下回的TBS治疗　有研究认为PSA患者语言功能恢复的关键可能在于左半球语言功能残留区的激活[37]。其中额下回(inferior frontal gyrus,IFG)是一个重要的区域,涉及各种任务,包括语言理解、言语产生、语义处理、精细运动控制、感受意识和情绪[38]。此外,额下回参与语言连贯性的控制,帮助说话者组织语言,使言语表达更加流畅和连贯。SZAFLARSKI等[39]以功能磁共振成像研究(functional magnetic resonance imaging,fMRI)为引导,将iTBS作用于左额叶残余语言皮层即左额下回和附近区域,发现可以改善患者语义流畅度,治疗后患者右半球激活减少和靠近刺激部位的左额叶语言偏侧化增加,这种从右半球到左半球激活的转变为TBS促进皮质重组提供了部分证据。另有研究提出靶点精准选择的重要性,当以fMRI检查中激活最显著的左额下回及周围区域为iTBS刺激靶点时,PSA患者命名能力和语义流畅性在治疗后立即改善,并持续至少3个月[40]。GRIFFIS等[41]研究中患者动词生成能力的改善同样验证了左额下回作为iTBS刺激靶点的有效性,虽然缺乏对照组,但MRI数据证明了从右到左的语言偏侧化,右额下回激活的变化与流畅度的提高呈负相关关系,iTBS刺激残余左半球语言区域可能会减少与语言产生相关的对侧反应并促进卒中后残余语言区域的募集。CHOU等[42]将iTBS作用在慢性PSA患者左Broca区,患者整体语言表现方面取得了更好的结果,还提高了听理解能力。虽然左额下回主要与语言表达有关,但左额下回的损伤也可能影响语言的理解能力和复述能力,尤其是在复杂句子结构的理解上。左额下回与其他大脑区域的连接对于语言功能也是必要的,例如与Wernicke区(语言理解的关键区域)的连接影响语言的理解和产生。

言语训练结合TBS联合治疗在一定程度上有益于PSA的恢复。ALLENDORFER等[43]研究发现,患者在完成iTBS刺激左额下回治疗后,接着进行45～60 min的限制诱导失语症治疗(constraint-induced aphasia therapy,CIAT)能提升命名能力,但是此研究样本量相对较少,且没有设置对照组,暂时无法比较TBS和CIAT发挥作用各占比。

综上,iTBS刺激左额下回可提高PSA患者言语流畅度、命名及听理解能力,恢复过程中涉及左半球残留区的再募集及从右半球到左半球的语言偏侧化,上述研究表明左额下回的功能重组和网络连接的恢复对于语言功能的改善至关重要。

2.1.2　左额上回的TBS治疗　曾有研究表明,在言语恢复过程中左额上回(superior frontal gyrus,SFG)与额下回的功能连接增加[44]。尽管额上回不直接参与语言的产生,但通过与大脑其他语言区域如Broca区和Wernicke区的连接,影响语言的流畅性和连贯性,该区还参与手势和言语的整合,影响多模态交流。REN等[17]研究表明,左额上回iTBS治疗和右额上回cTBS治疗皆可改善患者言语能力,额上回与传统的刺激部位额下回相比更容易靶向刺激,因为该区域组织在左右大脑半球通常未受损,而在许多PSA患者中额下回和周围组织存在病变。上述研究认为左额上回可能是TBS治疗方案的一个潜在靶点,在PSA患者中是安全且可耐受的。

2.1.3　左额叶背外侧的TBS治疗　对1例之前从未接受过言语治疗的慢性左半球缺血性卒中的老年男性患者进行了iTBS治疗,刺激部位为左额叶背外侧,这个区域在认知功能中扮演着关键角色,是执行控制功能的关键区域,包括任务转换、预防干扰、抑制控制、规划和工作记忆等,这些功能对于语言的流畅交流至关重要[45-47]。左额叶背外侧在处理语

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义时发挥作用，特别是在处理讽刺、挖苦等需要情感和社会认知加工的语言形式时活跃。

治疗前，该患者被诊断为轻度表达性失语症，伴有短期记忆障碍和语言信息处理障碍。在进行iTBS治疗的同时，患者还接受了计算机辅助的工作记忆训练，研究结果显示，患者的工作记忆、推理能力、叙事能力、沟通效率和生活质量均有显著改善，而且在言语功能的亚项中，命名和阅读能力也得到了提升，该研究提示左额叶背外侧是iTBS治疗PSA的潜在靶区，通过调节其活动可能改善语言功能。

2.1.4　左初级运动皮层的TBS治疗　初级运动皮层即M1区比Broca区等经典语言区域更容易定位，目前有一种“一石二鸟”的潜在实用方法——刺激M1区用于治疗脑卒中患者的语言和运动障碍。1项初步研究表明，M1区通过结构和/或功能连接与语言系统相连，左M1区iTBS可能会诱导PSA患者语义系统的功能连接（functional connectivity，FC）变化，FC反映远隔脑区在功能活动上的相互作用[27]。XU等[48]发现iTBS干预PSA患者左M1区后，右额叶和左顶叶低频震荡振幅值（fractional amplitude of low frequency fluctuations，fALFF）降低，fALFF值反映局部脑自发神经活动，此治疗方案可抑制右侧内侧额叶的脑活动，同侧半球顶叶fALFF值下降，表明iTBS的直接效果可能不会改善认知相关顶叶区域脑神经元的兴奋性。

综上，基于半球间抑制失衡理论模型，TBS可能通过调节左右半球间的相互作用来促进PSA患者言语功能恢复。

2.1.5　左Wernicke区的TBS治疗　左Wernicke区是语言理解的核心，负责处理语音和形成语义，帮助理解句子意义，参与词汇选择和语言检索，与语言记忆紧密相关，尤其是单词的听觉记忆。虽然不直接产生语言，但通过与Broca区的连接来保证语言流畅，该区受损时可能导致感觉性失语症，表现为患者言语无意义，难以理解他人。该区域还与听觉皮层相连，将听觉信号转化为语言，并与Broca区共同协调语言功能。

VERSACE等[49]将iTBS以随机顺序应用于慢性流利性PSA患者的Wernicke区、Wernicke区的右半球同源区和初级视觉皮层，在左Wernicke区进行iTBS后的5 min，检测到听理解能力显著提高，这种影响随后减弱，40 min后消失，未在右半球同源区和初级视觉皮层发现高度特异性变化，仍需要进一步的研究来明确TBS诱导的调节是否可以通过重复的TBS疗程以及将TBS与言语治疗相结合来增强并转化为更持久的效果。

2.2　TBS作用于PSA患者右侧大脑半球的研究

尽管大多数TBS研究集中在PSA患者左侧大脑半球，但右侧大脑半球在PSA恢复过程中的作用同样不容忽视。近年来，研究者开始将TBS应用于PSA患者右侧大脑半球，探讨TBS对PSA患者右侧大脑半球的影响，旨在揭示其在语言功能恢复中的潜在作用机制，为失语症的临床治疗提供新的策略和理论支持。研究表明，针对右半球特定脑区进行TBS刺激可以促进语言功能的恢复，但不同靶点的刺激疗效存在差异。与大部分研究集中于刺激左额下回相似，并且右额下回尤其受到关注。

2.2.1　右额下回的TBS治疗　既往研究表明，使用针对右额下回的抑制性刺激有助于促进PSA患者的康复[50]。虽然在PSA患者左侧大脑半球进行高频刺激效果良好，但在右侧大脑半球上使用低频刺激治疗的效应更长。GEORGIOU等[51]发现选择右额下回后部（posterior inferior frontal gyri，pIFG）作为cTBS刺激靶点，患者有自发表达、命名和阅读改善的趋势，且有随访患者表示其理解和阅读能力的改善得到了长期维持。该研究建议将右额下回后部的抑制性TMS作为一种促进PSA患者语言恢复的独立治疗方法。额下回后部影响言语表达流畅性，对该区的抑制可随神经可塑性伸展方向，影响整个言语网络。

有研究人员在PSA患者右额下回三角部（pars triangularis，PTr）施以cTBS，通过促进单词检索期间的语音访问来增强命名，表明命名障碍局限于此处理阶段的个体可能最有可能受益[52]，GEORGIOU等[53]研究结果显示，cTBS治疗后患者听理解、言语表达能力有所改善，生活质量也有所改善。也有研究显示，iTBS结合旋律语调疗法治疗后，患者语言流利度和短语重复性有所改善[54]。

上述研究表明，将cTBS作用于PSA患者的右额下回可能改善PSA患者自发表达、命名和阅读能力，结合患者实际失语情况可选择该区作为重要刺激靶点。

2.2.2　右颞上回后部的TBS治疗　ZHENG等[55]分别对患者Wernicke区同源的区域即右颞上回后部（posterior superior temporal gyrus，pSTG）进行cTBS或假刺激，然后进行30 min的言语训练，结果显示

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cTBS组的失语商、听理解、复述能力高于假刺激组,结合fMRI研究结果表明,右颞上回后部的cTBS可能通过抑制右额叶-丘脑-小脑回路的内在活性和增强右颞顶区的参与来改善语言产生。

2.2.3 右唇部运动皮层的TBS治疗　有研究认为运动皮层有助于PSA的恢复,它参与了单词的产生,该研究针对患有语音缺陷的PSA患者的右唇部运动皮层使用cTBS,发现可以改善命名能力[56]。

2.3 TBS作用于PSA患者多个靶点的研究

如上所述,虽然TBS刺激单一靶点可以改善PSA患者的某项或多项言语功能,但单一靶点刺激很难改善PSA患者的所有言语亚项功能。先进的影像学技术发现PSA患者参与语言功能恢复的脑区不止一个,实际上PSA患者的失语类型及临床表现也互不相同,仅刺激单个特定脑区难以改善或相同程度地改善PSA患者所有言语亚项功能。这种差异性提示临床人员在制定治疗计划时,为达到精准康复治疗,可根据临床研究并结合患者实际具体临床症状和损伤部位选择合适的多靶点刺激方案。近年来部分双侧多靶点刺激研究表示其对PSA患者言语功能恢复有利。

2.3.1 双侧半球的多靶点TBS治疗　KHEDR等[57]首次进行双侧半球多靶点rTMS刺激,实验组采用右Broca区低频rTMS刺激及左Broca区高频rTMS刺激并联合言语训练的方案,连续治疗10 d后实验组患者理解、命名、重复和流畅度功能较对照组改善明显,该研究除了试图消除从非优势半球对优势半球的抑制作用外,还试图加强代谢功能障碍区域内的神经连接。

VUKSANOVIĆ等[58]报告了1例慢性卒中后非流利失语症患者左Broca区iTBS与右侧同源区cTBS相结合的研究,结果显示该患者命题言语、语义流畅性、短期语言记忆和语言学习显著改善。闫芳等[59]采用右Broca区先于左Broca区的序贯rTMS刺激方案,与接受常规言语的对照组相比,双侧半球多靶点刺激组患者言语改善更为明显。后续研究或许可考虑调整TBS刺激顺序,如左侧优先,或探索双侧同时刺激模式,以系统评估刺激时序对PSA疗效的潜在影响。

2.3.2 单侧半球的多靶点TBS治疗　也有团队正在开展单侧多靶点刺激方案治疗PSA研究。HUANG等[60]计划采用cTBS模式依次刺激PSA患者右额下回、颞上回、额上回,并结合言语训练,该方案可能帮助言语功能达到较为全面的恢复效果。

TBS的治疗过程极为迅速,即便是对多个脑区进行刺激,其所需的时间也远远少于传统TMS治疗,这种时间效率的显著提升,使得TBS多靶点刺激方案在临床应用中具有更高的可行性。单靶点刺激的治疗时间较短,但其对改善言语功能亚项的效果相对有限。相比之下,多靶点刺激的治疗时间较长,但能够广泛地影响神经网络,在改善言语功能亚项上实现了各靶点之间的互补效果。大脑中动脉完全闭塞的患者不适合接受双侧半球刺激,由于左Broca区受损严重,更适合增强右同源区的激活,应根据患者的实际言语功能状况和时间安排来选择合适的治疗方案[57]。

3 小结与展望

已有的研究显示,iTBS刺激左额下回、左Wernicke区、左额叶背外侧以及cTBS刺激右额下回后部、右额下回三角部、右颞上回后部及右唇部运动皮层均可改善PSA患者的言语功能,但不同的刺激部位对患者语言功能亚项的影响不尽相同。iTBS刺激左额下回在改善PSA患者整体言语功能的同时对命名和语义流畅度的改善相对更加明显,刺激左Wernicke区可观察到对听理解能力的促进,而刺激左额叶背外侧则对命名和阅读能力有一定的正面影响。cTBS刺激右额下回后部有助于提升PSA患者的自发表达、命名和阅读能力,刺激右额下回三角部可增强命名能力、听理解和言语表达能力,刺激右颞上回后部则侧重改善听理解和复述能力,刺激右唇部运动皮层可能对命名能力有所提升。因此,在临床上治疗PSA患者时,应针对不同类型的PSA患者以及PSA患者的不同失语表现选择不同的刺激靶点,精准改善患者对应的言语功能,既可采取多靶点的互补作用,又可选择多靶点的叠加作用。值得重视的是,TBS刺激治疗时间短,在多靶点刺激上具有操作上的优势,更具有可行性。

除了本文提到的刺激靶点,影像学技术的进步使得越来越多的脑区被发现可能参与语言的功能活动。有研究者认为,小脑在恢复PSA患者言语功能、调节大脑半球语言网络和语言学习机制方面具有重要作用[61-62]。张晓彤等[63]研究发现,PSA患者右小脑Crus Ⅰ、Ⅱ分别对左额中回、左额下回三角部的有效连接增强,右小脑可能是PSA患者促进语言恢复的潜在调控位点。SEBASTIAN等[64]和

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ZHENG等[65]认为对于左半球具有不同病变部位和大小的失语症人群来说右小脑可能是促进PSA患者言语功能康复的最佳刺激部位。因此，今后PSA患者的TBS治疗还需要要寻找更多的刺激靶点，并根据患者的语言功能选择相应的刺激靶点及/或多靶点治疗。

此外，现有的研究局限于探讨TBS刺激PSA患者言语功能改善的短期效果，缺乏长期效果的临床数据，今后的研究可以使用西方失语症成套测验、脑卒中失语症生活质量量表等专业的神经心理学量表，继续跟踪患者治疗后半年或者1年的语言功能评分、生活质量评分等指标的变化来评估患者长期疗效，结合功能磁共振成像、近红外脑功能成像等影像学技术观察PSA患者脑结构及脑功能的变化，进一步探讨TBS治疗PSA的脑机制，为PSA的精准治疗提供理论基础。

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Effect of *Theta* Burst Transcranial Magnetic Stimulation on Post-Stroke Aphasia Based on Different Stimulation Targets in the Cerebral Cortex

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ABSTRACT Currently, the population affected by post-stroke aphasia (PSA) is increasing steadily and clinically these patients present with speech impairments of varying degrees. As a unique core cognitive and social function specific to humans, speech ability is crucial across multiple dimensions, including physical and mental health as well as cultural inheritance. However, the cerebral mechanisms underlying the onset and recovery of PSA remain unclear, and traditional treatments yield suboptimal results, prompting researchers to continuously explore new therapeutic approaches. Transcranial magnetic stimulation (TMS) is a non-invasive, safe brain stimulation technique. *Theta* burst transcranial magnetic stimulation (TBS), a specialized TMS mode, offers advantages such as short treatment duration, low intensity, strong efficacy, and minimal side effects, leading to its growing clinical application in recent years. This article reviews the characteristics, common parameters, and safety of TBS; notably, therapeutic effects on PSA patients vary with different TBS parameters, and more research is needed to identify the optimal parameters. Existing studies suggest that the

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therapeutic mechanisms for PSA include regulating cerebral metabolism, improving functional network connectivity, and promoting interhemispheric excitatory balance. Current domestic and international research results indicate that when TBS acts on cerebral cortical regions, such as the inferior frontal gyrus, primary motor cortex, Wernicke's area, and dorsolateral prefrontal cortex, it can improve one or more of the subdomains of speech function, including spontaneous speech, auditory comprehension, repetition, and naming, which suggests that TBS has potential clinical significance for the recovery of speech function in PSA patients. In addition, therapeutic regimens combining bilateral or unilateral multi-target TBS stimulation with speech training can achieve functional complementarity, and due to the short duration of TBS, multi-target stimulation is highly feasible. Currently, research has limitations such as focusing mainly on short-term effects and lacking of long-term data. In the future, it is necessary to explore more potential targets such as the cerebellum, combine neuropsychological scales with imaging techniques to evaluate long-term efficacy and further explore cerebral mechanisms, so as to provide a basis for precise treatment of PSA.
KEY WORDS　*Theta* burst transcranial magnetic stimulation; post-stroke aphasia; stimulation target; therapeutic efficacy;speech function
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Effect of Ultrasound-Guided Nerve Hydrodissection Combined with Rehabilitation Manual Therapy on Patients with Cubital Tunnel Syndrome with Long Course

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ABSTRACT　**Objective**　To explore new clinical diagnosis and treatment strategies and technical approaches for cubital tunnel syndrome with a long disease course by treating one patient with cubital tunnel syndrome duration exceeding 35 years using nerve hydrodissection combined with rehabilitative manual therapy. **Methods**　The rehabilitation treatment plan for this patient included: 1) Treatment of the right elbow using laser therapy, infrared therapy, and intermediate-frequency pulsed electrotherapy, once daily for 20 minutes, over a period of 3 weeks; 2) Application of Maitland Grade Ⅲ and Ⅳ joint mobilization techniques to improve joint range of motion, once daily for 10 minutes, for 3 weeks; 3) Gentle stretching of the ulnar nerve through specific movements such as shoulder abduction, elbow extension, and wrist extension to enhance nerve gliding, once daily for 10 minutes, over a period of 3 weeks; 4) Nerve hydrodissection under ultrasound guidance, once weekly, for a total of 3 treatments over 3 weeks. The changes of the patient before and after treatment were evaluated using muscle strength test, active range of motion (AROM) measurement, Action Research Arm Test (ARAT), and Numeric Rating Scale (NRS) for pain. **Results**　The abductor muscle strength of the right little finger was grade 4 before treatment, and grade 5^- after treatment. There was no significant difference in the deep flexor muscle strength of the ring and the little finger, muscle strength for finger abduction/adduction, opposition muscle strength of the little finger, and the adduction muscle strength of the thumb before and after treatment.The AROM for right elbow extension was limited to 25° before treatment and improved to 20° after treatment, with no significant change in elbow flexion. The NRS score for pain in the ulnar side of the right forearm and the ring and little fingers was 8 points before treatment, and reduced to 4 points after treatment, indicating an improvement from severe pain to moderate pain. The ARAT score was 46 points before treatment, and increased to 50 points after treatment. **Conclusion**　For cubital tunnel syndrome with an extended disease course, surgical treatment is generally preferred. However, this case was treated with ultrasound-guided nerve hydrodissection, achieving favorable clinical results. This provides a new clinical approach and technical means for the non-surgical treatment of cubital tunnel syndrome with an prolonged disease course.
KEY WORDS　cubital tunnel syndrome; ulnar nerve compression; nerve hydrodissection; ultrasound guidance; prolonged disease course
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